

Which Derivative Source for SINDy on Noisy Lorenz-63? A Small CPU Comparison of Finite Differences, Smoothing, TV and the Weak Form

Beryl-17H

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Abstract—Sparse identification of nonlinear dynamics (SINDy) needs time derivatives that noisy measurements do not provide. We compare five ways of obtaining them, with an identical degree-2 library and a sequentially thresholded least squares (STLSQ) back end, on simulated Lorenz-63 at six noise levels (0–10% of the signal standard deviation, 20 trajectories per level): finite differences, Savitzky–Golay, a smoothing spline, total-variation (TV) regularised differentiation, and a weak (integral) form that avoids differentiating the data. All five are reimplemented, with smoothing parameters and thresholds tuned per noise level on five separate trajectories. The hypothesis that the weak form recovers the exact support more often than every derivative-based method at every noise level of at least 1% is refuted: at 1% it ties three baselines at 1.00, at 2% (1.00 against 0.80–0.90) and at 5% (0.65 against 0.00–0.40) its lead is statistically significant against only some baselines, and at 10% no system exceeds 0.10. The weak form has the lowest mean coefficient error at every level, significantly against all four baselines only at 0% and 10%. The hypothesis that smoothing lowers the coefficient error of finite differences is refuted by the registered test: at 5% all three smoothers have a higher mean error than finite differences. This verdict depends on the finite-difference threshold, which the tuner’s tie rule fixed after a first round of results; at equal thresholds only Savitzky–Golay is above finite differences throughout, and TV is below. A threshold grid on the test trajectories shows that up to 1% noise every method recovers every support at a threshold of 0.4, and that the low-noise ordering of the methods reversed when the tie rule was changed; at 5% the weak form’s rate is above the best threshold of each baseline (0.65 against at most 0.40). This is a one-system, 20-trial CPU study with oracle-assisted tuning.

Index Terms—equation discovery, SINDy, STLSQ, numerical differentiation, weak form, Lorenz-63, noise robustness

I. Introduction

Sparse regression on a library of candidate terms is a popular route to recovering governing equations from data. SINDy [1] regresses the time derivative of the state on a library and promotes sparsity with sequentially thresholded least squares; PDE-FIND [2] extended the idea to partial differential equations. The derivative is not

measured, so it has to be estimated from noisy samples, and the quality of that estimate often decides whether the correct terms are found. Toolboxes such as PySINDy [3] therefore expose several differentiators, and the weak or integral formulation [4]–[7] sidesteps differentiation altogether by integrating the dynamics against test functions.

The ODE Weak SINDy paper [5] compares the weak form with standard SINDy. This study asks a narrower, controlled question about all of these options: with the same library, the same STLSQ solver, the same data and the same tuning protocol, how do finite differences, Savitzky–Golay [8], a smoothing spline, TV-regularised differentiation [9] and the weak form compare in the rate of exact support recovery and in coefficient error on Lorenz-63 [10] as the noise grows from 0 to 10%?

We registered five hypotheses before the runs (Sec. IV). The outcome is mostly negative: H1 (the weak form beats every derivative-based method in support rate from 1% noise) and H2 (every smoother has a lower coefficient error than finite differences) are refuted by the registered tests, H2 with a verdict that depends on the finite-difference threshold at 5%; H3 (all systems are exact at zero noise) is supported; H4 (too narrow a test-function window loses the advantage) is partly supported, since the width matters but in the opposite direction; H5 (a library on the smoothed state improves support recovery) is not supported. Contributions are (i) a controlled, reproducible comparison with failures split into false positives and negatives; (ii) a grid of every system at every STLSQ threshold, which shows that the low-noise ordering of the methods is decided by threshold selection, including the tie rule of our own tuner, revised after a first round of results; (iii) width, exponent and library-state ablations. Scale is small: one simulated system, 20 trajectories per cell, reimplemented methods, CPU.

II. Related Work

SINDy and relatives. SINDy [1] and its PDE counterpart PDE-FIND [2] use sparse regression over a candidate library. SINDy-PI [11] targets implicit dynamics with a parallel formulation reported to be robust to noise, and

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Ensemble-SINDy [12] bags models for the low-data, high-noise regime; neither is evaluated here. PySINDy [3] packages these methods with several differentiation options.

Numerical differentiation of noisy data. Savitzky–Golay local polynomial fitting [8] and TV-regularised differentiation [9], which allows non-smooth derivatives, are the classical smoothing differentiators. van Breugel et al. [13] cast the choice of method and parameters as a multi-objective optimisation and stress that parameter selection is the practical difficulty.

Weak and integral formulations. Schaeffer and McCalla [4] identify dynamics from integral terms, Reinbold et al. [7] use a weak form to discover models from noisy and incomplete spatiotemporal data, and Weak SINDy integrates the dynamics against test functions so that pointwise derivative approximations are replaced by linear transformations of the data, for ordinary [5] and partial [6] differential equations. The ODE paper [5] is the direct precedent of this study: it reports that the weak form improves on standard SINDy by orders of magnitude and identifies the correct terms in both small- and large-noise regimes.

Gap. Weak SINDy [5] is compared with standard SINDy in its own paper, and the differentiation literature [9], [13] evaluates derivative estimates, not the equations recovered from them. We did not find a head-to-head of all five derivative sources with a shared STLSQ back end, one tuning protocol and the exact-support rate as the headline metric; this study provides one at small scale. Our weak form is far simpler than WSINDy (one test-function family, fixed width, no reweighting), so the results do not bear on that algorithm as published.

III. Method

Model and library. For a state $\mathbf{x}(t) \in \mathbb{R}^3$ with $\dot{\mathbf{x}} = \Theta(\mathbf{x})\Xi$, $\Theta(\mathbf{x}) = [1, x, y, z, x^2, xy, xz, y^2, yz, z^2]$ and $\Xi \in \mathbb{R}^{10 \times 3}$ has 7 nonzero entries for Lorenz-63 ($\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$; $\sigma = 10, \rho = 28, \beta = 8/3$). Given noisy samples $\tilde{\mathbf{x}}_i = \mathbf{x}(t_i) + \boldsymbol{\varepsilon}_i$, every method builds a regression problem (A, B) with $A \in \mathbb{R}^{m \times 10}$, $B \in \mathbb{R}^{m \times 3}$ and solves $\min_{\Xi} \|A\Xi - B\|_F^2$ under a sparsity constraint.

STLSQ. Columns of A are normalised to unit norm. Start from the least-squares solution; repeat 10 times: set entries with $|\xi_{jk}| < \lambda$ (in original coefficient units) to zero and refit each equation k by least squares on its surviving columns.

Derivative-based systems (reimplemented): $A = \Theta(\tilde{\mathbf{x}})$ on the raw noisy state, $B = \dot{\tilde{\mathbf{x}}}$, the first and last 5% of samples removed. The derivative estimators are: (i) FD-STLSQ, second-order differences $\hat{x}_i = (\tilde{x}_{i+1} - \tilde{x}_{i-1})/2\Delta t$ (one-sided second-order at the ends); (ii) SG-STLSQ, the derivative of the local cubic polynomial fitted in a window of w samples [8]; (iii) Spline-STLSQ, the derivative of the cubic smoothing spline $s = \arg \min_s \sum_i (\tilde{x}_i - s(t_i))^2 + \mu \int s''(t)^2 dt$ with penalty μ ; (iv) TV-STLSQ, Chartrand’s regularised differentiation [9]: $u = \arg \min_u \alpha \text{TV}(u) +$

$\frac{1}{2} \|\mathcal{A}u - (\tilde{x} - \tilde{x}_0)\|^2$ with $\mathcal{A}$ the trapezoid cumulative integral, solved by 20 lagged-diffusivity Newton iterations ($\epsilon = 10^{-6}$).

Weak-form STLSQ (method). For compactly supported $\phi_k(t) = (1 - s^2)^p$, $s = (t - c_k)/(w/2)$, on $K = 200$ equispaced window centres c_k (windows fully inside the record), integration by parts with $\phi_k = 0$ at the window ends gives

$$\int \phi_k \dot{\mathbf{x}} dt = - \int \dot{\phi}_k \mathbf{x} dt \approx \left(\int \phi_k \Theta(\mathbf{x}) dt \right) \Xi, \quad (1)$$

so $A_{kj} = \sum_i w_i \phi_k(t_i) \Theta_j(\tilde{\mathbf{x}}_i)$ and $B_{k\cdot} = - \sum_i w_i \dot{\phi}_k(t_i) \tilde{\mathbf{x}}_i$ with trapezoid weights w_i ; $p = 3$ and the width w is tuned. No derivative of the data is taken, but the library is still evaluated on noisy states (errors-in-variables bias remains).

Metrics. With $\hat{\Xi}$ the STLSQ output: `support_exact` = $\mathbb{1}[\text{supp } \hat{\Xi} = \text{supp } \Xi]$ over all 30 entries; `coef_err` = $\|\hat{\Xi} - \Xi\|_F / \|\Xi\|_F$; `coef_err_oracle` is the same after refitting least squares on the true support with the same (A, B) , which isolates the quality of the regression problem from the selection step; false positives (FP) and false negatives (FN) are the numbers of spurious and missed entries.

IV. Experimental Setup

Data. All data are simulated. Lorenz-63 is integrated with DOP853 (rtol and atol 10^{-12}) from $(-8, 7, 27) + \mathcal{N}(0, I)$ after a 5-time-unit burn-in, then sampled for 10 time units at $\Delta t = 0.01$ (1001 samples). Gaussian noise with standard deviation equal to a fraction of the per-component standard deviation of the clean signal is added at levels 0, 0.5, 1, 2, 5, 10%. Each seed (0–19) gives a new initial condition and noise draw; every system sees the same 20 trajectories per level (6 levels $\times$ 5 systems $\times$ 20 seeds = 600 main runs).

Tuning. For each system and noise level, the smoothing parameter and the STLSQ threshold λ are chosen by grid search on 5 separate trajectories (seeds 1000–1004): highest exact-support rate, then lowest median coefficient error, then the largest threshold. A tie in the tuner is therefore equality of the support rate and of the median error on the five trajectories; it does not require identical models. The grids are: $\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.6\}$; SG $w \in \{11, 21, 41, 81, 161\}$; spline $\mu \in \{10^{-9}, \dots, 10^{-3}\}$ (7 values); TV $\alpha \in \{10^{-5}, \dots, 10^{-1}\}$; weak $w \in \{0.3, 0.6, 1, 1.5, 2.5\}$ time units. Tuning uses the ground-truth coefficients (oracle-assisted, equal for all systems), so absolute rates are optimistic. Selected values often lie on a grid edge (SG $w = 11$ at five of six levels; $\lambda = 0.6$ in 20 of the 30 cells of Table I). In five cells (FD at 5%; FD, SG, spline and TV at 10%) no configuration recovered the support of any tuning trajectory, and λ was selected by the median error alone. All systems share the same data, metric code and defaults; no baseline received other settings.

TABLE I

Tuned threshold λ (in parentheses the value under the first tie rule where it differs); last row: mean seconds per trial over the 120 main runs of the system.

Noise	FD	SG	Spline	TV	Weak
0%	0.6 (0.05)	0.6 (0.05)	0.6 (0.05)	0.6 (0.05)	0.6 (0.05)
0.5%	0.6 (0.1)	0.6 (0.1)	0.6 (0.1)	0.6 (0.1)	0.6 (0.05)
1%	0.6 (0.2)	0.6 (0.2)	0.6 (0.2)	0.6 (0.2)	0.6 (0.1)
2%	0.4	0.4	0.4	0.4	0.6 (0.2)
5%	0.1 (0.05)	0.4	0.4	0.4	0.6
10%	0.6 (0.4)	0.2	0.6 (0.4)	0.2	0.6
s/trial	0.10	0.10	0.12	1.46	0.10

Revision of the tie rule. The first version of the tuner resolved ties by grid order, i.e. by the smallest threshold. Ties decided λ in 19 of the 30 cells, and an audit showed that the conclusions below 5% noise and for FD at 5% depended on this arbitrary choice. The rule was replaced by the largest-threshold rule above (same tuning seeds and grids, no test data, smoothing parameters unchanged), and the main runs, the width sweep and the exponent ablation were run again. The new rule was adopted after the first test results were known, so it is not pre-registered; Table I lists both sets of thresholds, Table V reports every system at every threshold with both choices marked, and the first-version runs stay in the registry as superseded.

What a tie is. A diagnostic on the tuning seeds (method/tie_check.py, one run per noise level) compared the models of the tied configurations. In 17 of the 19 tie cells they are identical on all five trajectories. In the other two the tied thresholds agree only in rate and median: for FD at 5% (0.05 and 0.1), the cell on which the H2 verdict turns, the models differ on 3 of the 5 trajectories, and for spline at 10% (0.4 and 0.6) on 1 of 5. In both, the selected threshold also has the lower mean tuning error, and a tie-break by mean error would select the same configuration in all 30 cells; the FD threshold at 5% is nonetheless a choice between different models made on five trajectories.

Hypotheses and tests (wording of proposal.md, written before the runs; tests are Welch’s t -test from `rh.compare`, $p < 0.05$, no multiple-comparison correction). H1: at every noise level $\geq 1\%$ the weak form has a higher exact-support rate than every derivative-based baseline (higher mean and $p < 0.05$). H2: SG, spline and TV each have a lower mean coefficient error than FD at noise $\geq 1\%$; refuted if any smoother is not lower at some level. H3: at 0% noise all five systems recover the support in all seeds. H4: the weak-form advantage depends on the test-function width, and too narrow a window loses the advantage (descriptive width sweep at 2% noise). H5: evaluating the library on the smoothed state (SG, spline) instead of the raw noisy state improves support recovery at 2% noise (descriptive).

Compute. Apple M1 Pro CPU, 2 threads; the runtime per trial (Table I) includes data generation. The registry holds 1547 current runs (600 main, 600 threshold-grid,

200 width, 60 exponent and 80 library-state runs, a smoke test and six tie-diagnostic runs) among 2648 logged runs; the others are superseded (the first-version runs and one earlier tie diagnostic); wall-clock was about 30 minutes for the first version and 20 for the revision.

V. Results

H1 (weak form has a higher support rate than every baseline at every level from 1%): refuted. At 1% the weak form, SG, spline and TV all recover every support (Table IIa), so there is no difference to test. At 2% the weak form is at 1.00 against 0.80–0.90; the difference is significant against FD and spline ($p = 0.042$, uncorrected) but not against SG or TV (Table IIIa). At 5% it is at 0.65 against 0.00–0.40, significant against FD, SG and spline but not against TV ($p = 0.119$). At 10% every system fails (weak 0.10, others 0.00, not significant). The weak form is never below a baseline, but the significant win over all four baselines that H1 requires holds at no level. The picture below 5% depends on how the threshold is chosen (Sec. VI): with the first tie rule the weak form was at 0.75 at 0.5, 1 and 2%, below baselines at 0.80–0.90.

Where the failures come from. Table IV separates spurious terms (FP) from missed terms (FN). Up to 2% the weak form has neither. The baselines at 2% both add (0.10–0.20 per trial) and miss (0.05–0.10) terms. At 5% the baselines mostly add terms (0.65–2.70 FP against 0.00–0.25 FN), while the weak form has 0.20 FP and 0.25 FN. FD at 5% never recovers the support and carries 2.70 spurious terms per trial at its threshold of 0.1, picked by median error alone (Sec. IV); larger thresholds do not give a high rate either (0.15 at 0.4 and 0.25 at 0.6, Table V).

H2 (each smoother has a lower mean coefficient error than FD from 1%): refuted. At 1, 2 and 10% the means of SG, spline and TV are below FD (Table IIb), significantly only for spline and TV at 1% (Table IIIc). At 5% all three are above FD (0.0536–0.0772 against 0.0469), significantly for SG and spline, which refutes H2 as registered. This verdict depends on the FD threshold: at the first-version value of 0.05 the FD error is 0.1354, above all three smoothers at their tuned thresholds (Table VI), and 0.05 and 0.1 were tied in the tuner although they give different models on three of the five tuning trajectories (Sec. IV). The claim for TV is the weakest: its higher mean is not significant (Table IIIc), and at equal thresholds TV is below FD in all five columns of Table VI. Spline is above FD at four of the five equal thresholds (not at 0.05) and SG at all five; on the true support FD is below SG and spline but not TV (0.0351 against 0.0655, 0.0472 and 0.0324, Table IIId).

H3 (all systems exact at 0% noise): supported. All five systems recover the support in all 20 trials at 0% and at 0.5%. Under the first tie rule ($\lambda = 0.05$) FD and SG were at 0.95, a threshold effect: both are at 1.00 for every $\lambda \geq 0.1$ (Table V).

TABLE II

Main results (the 600 main runs, 20 seeds per cell): (a) exact-support rate, (b) mean and (c) median relative coefficient error, (d) mean coefficient error when the regression is restricted to the true support (isolates the regression problem from support selection).

(a) exact-support rate						
System	0%	0.5%	1%	2%	5%	10%
FD	1.00	1.00	0.95	0.80	0.00	0.00
SG	1.00	1.00	1.00	0.90	0.25	0.00
Spline	1.00	1.00	1.00	0.80	0.20	0.00
TV	1.00	1.00	1.00	0.85	0.40	0.00
Weak	1.00	1.00	1.00	1.00	0.65	0.10

(b) mean coefficient error						
System	0%	0.5%	1%	2%	5%	10%
FD	0.0065	0.0068	0.0118	0.0201	0.0469	0.3013
SG	0.0033	0.0037	0.0046	0.0126	0.0772	0.1721
Spline	1.5e-5	8.9e-4	0.0021	0.0150	0.0636	0.2485
TV	0.0025	0.0022	0.0020	0.0095	0.0536	0.1473
Weak	1.0e-8	8.6e-4	0.0017	0.0035	0.0337	0.0941

(c) median coefficient error						
System	0%	0.5%	1%	2%	5%	10%
FD	0.0065	0.0071	0.0081	0.0129	0.0456	0.1337
SG	0.0034	0.0038	0.0046	0.0084	0.0739	0.1271
Spline	1.5e-5	7.7e-4	0.0019	0.0067	0.0571	0.1269
TV	0.0025	0.0021	0.0016	0.0046	0.0470	0.1224
Weak	1.1e-8	6.7e-4	0.0014	0.0027	0.0129	0.1094

(d) mean coefficient error on the true support						
System	0%	0.5%	1%	2%	5%	10%
FD	0.0065	0.0068	0.0078	0.0114	0.0351	0.1052
SG	0.0033	0.0037	0.0046	0.0082	0.0655	0.0988
Spline	1.5e-5	8.9e-4	0.0021	0.0058	0.0472	0.0987
TV	0.0025	0.0022	0.0020	0.0040	0.0324	0.0993
Weak	1.0e-8	8.6e-4	0.0017	0.0035	0.0094	0.0403

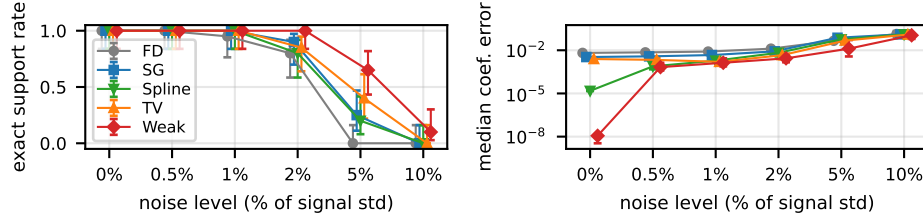


Fig. 1. Exact-support rate (95% Wilson intervals) and median coefficient error (interquartile range) versus noise level.

TABLE III

Welch p -values (*: $p < 0.05$; -: identical rates). In (a) no baseline has a higher rate and in (b) the weak form has the lower mean in every cell; in (c) $\ddagger$ marks a smoother whose mean is higher than FD's.

System	0%	0.5%	1%	2%	5%	10%
(a) support rate, weak form vs.						
FD	—	—	0.330	0.042*	0.000*	0.163
SG	—	—	—	0.163	0.010*	0.163
Spline	—	—	—	0.042*	0.003*	0.163
TV	—	—	—	0.083	0.119	0.163
(b) coefficient error, weak form vs.						
FD	0.000*	0.000*	0.024*	0.008*	0.179	0.037*
SG	0.000*	0.000*	0.000*	0.040*	0.000*	0.004*
Spline	0.000*	0.855	0.287	0.055	0.007*	0.012*
TV	0.000*	0.000*	0.550	0.170	0.088	0.014*
(c) coefficient error, FD vs.						
SG	0.000*	0.000*	0.096	0.286	0.000* $\ddagger$	0.187
Spline	0.000*	0.000*	0.030*	0.526	0.004* $\ddagger$	0.626
TV	0.000*	0.000*	0.027*	0.137	0.369 $\ddagger$	0.116

Coefficient error. The weak form has the lowest mean coefficient error at every noise level (Table IIb) and the lowest error on the true support at every level (Table IIc; 0.0094 against 0.0324–0.0655 at 5%, 0.0403 against 0.0987–0.1052 at 10%). The differences in Table IIb are significant against all four baselines only at 0% and 10%

TABLE IV

Mean false positives / false negatives per trial (30 coefficients, 7 nonzero).

Noise	FD	SG	Spline	TV	Weak
0%	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00
0.5%	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00
1%	0.00/0.05	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00
2%	0.20/0.10	0.10/0.05	0.15/0.10	0.15/0.05	0.00/0.00
5%	2.70/0.00	0.85/0.15	0.85/0.20	0.65/0.25	0.20/0.25
10%	1.45/1.20	2.85/0.80	1.30/1.25	2.60/0.70	0.75/0.85

(Table IIIb); against spline they are not significant from 0.5 to 2%, against TV not from 1 to 5%, and against FD not at 5%. At 10% the gap is much smaller in the medians (0.1094 against 0.1224–0.1337, Table IIc) than in the means (0.0941 against 0.1473–0.3013). The clearest separation is at 5%, where the weak-form median is 0.0129 against 0.0456–0.0739 (Fig. 1).

What the results do not show. The ordering of the systems below 5% noise is not resolved: it reverses with the tie rule of the tuner. Rate differences are poorly resolved at $n = 20$: 0.20 (1.00 against 0.80 at 2%) gives $p = 0.042$, while 0.15 at the same level ($p = 0.083$) and 0.25 at 5% (0.65 against 0.40, $p = 0.119$) are not significant. The p -values are uncorrected over the 18 to 24 comparisons of each table; the two $p = 0.042$ cells would not survive a

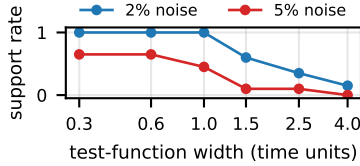


Fig. 2. Weak-form exact-support rate versus test-function width.

Bonferroni correction even over the four comparisons of that level.

VI. Ablations and Analysis

Threshold and tie rule. One run per system, level and seed builds the regression problem once and applies STLSQ with each threshold of the grid (Tables V and VI); the shaded cells reproduce the main runs exactly. (i) Up to 1% noise the derivative source makes no difference to support recovery at $\lambda = 0.4$ (all five systems at 1.00) and almost none at $\lambda = 0.6$ (only FD at 1% drops, to 0.95). At 1% smaller thresholds lower the rate of every system, down to 0.15–0.30 at $\lambda = 0.05$. The thresholds tied at these levels returned identical models on the tuning trajectories (Sec. IV), so at 0.5 and 1% the first tie rule produced the underlined rates of 0.75–0.90, with the weak form below the baselines, and the revised rule 0.95–1.00 for all. Neither ordering reflects the derivative source. (ii) At 2% only the weak form improves up to $\lambda = 0.6$ (1.00); the baselines reach 0.80–0.90 at $\lambda = 0.4$ and do not improve at 0.6. At equal thresholds the weak form is at least as good as every baseline for $\lambda \geq 0.2$ at 2% and for $\lambda \geq 0.4$ at 5%, but not in every cell below (at 5% and $\lambda = 0.2$ it is at 0.10 against 0.05–0.15). (iii) At 5% the best cell of each baseline over the whole grid (FD 0.25, SG 0.25, spline 0.20, TV 0.40) stays below the weak form’s 0.65, so the weak form’s higher rate at 5% does not depend on how the baselines’ thresholds are selected within this grid (against TV the difference is not significant, Table IIIa); at 10% no cell exceeds 0.10.

Test-function width (H4). Table VII and Fig. 2 vary the width of the weak-form test functions. At 2% noise, the level registered for H4, the support rate is 1.00 for widths 0.3, 0.6 and 1.0 and falls to 0.60, 0.35 and 0.15 at 1.5, 2.5 and 4.0, below the baselines’ 0.80–0.90. At 5% (added to the registered sweep) the rate is 0.65 at 0.3 and 0.6, 0.45 at 1.0 and at most 0.10 beyond. H4 is partly supported: the width matters strongly, but not in the registered direction, since the narrowest window tested keeps the rate and the wide ones lose it. Windows narrower than 0.3 were not tested, and no experiment isolates why wide windows fail.

Test-function exponent. At 2% noise the exponents $p = 2, 3, 4, 6$ (60 runs beyond the main setting $p = 3$) all give a support rate of 1.00, with mean coefficient errors of 0.0039, 0.0035, 0.0033 and 0.0031 (differences not tested): no detectable effect on support recovery in this range.

Library on the smoothed state (H5). With the library evaluated on the smoothed state instead of the noisy measurements (SG and spline, Table VIII), H5, registered for support recovery at 2%, is not supported: the rate goes from 0.90 to 0.85 for SG and from 0.80 to 0.85 for spline, and does not change at 5%. The coefficient error is lower with the smoothed library in all four cells (at 5%, SG 0.0772 to 0.0339 and spline 0.0636 to 0.0387); no test was run on this difference.

VII. Limitations

All data are simulated. The study has one dynamical system (Lorenz-63), one sampling interval ($\Delta t = 0.01$), one trajectory length (10 time units), one library (degree-2 polynomials, which contains the true model) and 20 trajectories per cell, so rate differences up to about 0.25 are not reliably resolved (Sec. V) and the p -values are uncorrected. All five methods are reimplemented in numpy with small hyperparameter grids whose edge is selected in most cells, so a wider grid might change the ranking. Tuning is oracle-assisted, which inflates absolute rates and is not available in practice. Five tuning trajectories do not determine the threshold: ties in support rate and median error decided it in 19 of 30 cells (in two of them, FD at 5% and spline at 10%, between thresholds that give different models), the rule that breaks them was changed after the first test results were seen, and the ordering of the systems below 5% noise follows that rule. Noise is i.i.d. Gaussian with a level known to the tuner. The weak form uses one test-function family and window count; other weak-SINDy variants, ensembles and implicit forms were not compared. A full study would add systems, sampling rates, noise models, tuning without ground truth on a larger tuning set, and more trials per cell.

VIII. Conclusion

On simulated noisy Lorenz-63 with a fixed library and STLSQ, the registered claim that the weak form beats every derivative-based method in exact-support rate at every noise level from 1% is refuted: it ties three baselines at 1%, its leads at 2% and 5% are significant against some baselines only, and at 10% every method fails. The weak form has the lowest mean coefficient error at every level, significantly against all four baselines only at 0% and 10%; the clearest separation, at 5%, is in the median (0.0129 against 0.0456–0.0739), while the mean there is not significantly below FD or TV. At the tuned thresholds no smoother lowered the mean coefficient error of finite differences at 5%, which refutes the second claim as registered; that verdict depends on the finite-difference threshold, and at equal thresholds only Savitzky–Golay is above finite differences throughout while TV is below. Up to 1% noise the support rate is governed by the STLSQ threshold more than by the derivative source, and a tie rule in the tuner was enough to reverse the low-noise ordering; at 5% the weak form’s rate is above the best threshold of

TABLE V

Exact-support rate on the 20 test seeds for every STLSQ threshold λ of the tuning grid (smoothing parameters as tuned). Shaded: tuned threshold (the main runs). Underlined: threshold selected by the first tie rule. Diagnostic only: these cells use the test seeds.

λ	0% noise					0.5% noise					1% noise				
	0.05	0.1	0.2	0.4	0.6	0.05	0.1	0.2	0.4	0.6	0.05	0.1	0.2	0.4	0.6
FD	<u>0.95</u>	1.00	1.00	1.00	<u>1.00</u>	0.45	<u>0.90</u>	1.00	1.00	<u>1.00</u>	0.15	0.40	<u>0.85</u>	1.00	<u>0.95</u>
SG	<u>0.95</u>	1.00	1.00	1.00	<u>1.00</u>	0.35	<u>0.85</u>	1.00	1.00	<u>1.00</u>	0.30	0.50	<u>0.85</u>	1.00	<u>1.00</u>
Spline	<u>1.00</u>	1.00	1.00	1.00	<u>1.00</u>	0.45	<u>0.85</u>	1.00	1.00	<u>1.00</u>	0.25	0.60	<u>0.85</u>	1.00	<u>1.00</u>
TV	<u>1.00</u>	1.00	1.00	1.00	<u>1.00</u>	0.55	<u>0.90</u>	1.00	1.00	<u>1.00</u>	0.30	0.45	<u>0.85</u>	1.00	<u>1.00</u>
Weak	<u>1.00</u>	1.00	1.00	1.00	<u>1.00</u>	<u>0.75</u>	0.90	1.00	1.00	<u>1.00</u>	0.25	<u>0.75</u>	0.90	1.00	<u>1.00</u>

λ	2% noise					5% noise					10% noise				
	0.05	0.1	0.2	0.4	0.6	0.05	0.1	0.2	0.4	0.6	0.05	0.1	0.2	0.4	0.6
FD	0.05	0.25	0.60	<u>0.80</u>	0.70	<u>0.00</u>	<u>0.00</u>	0.05	0.15	0.25	0.00	0.00	0.00	0.00	<u>0.00</u>
SG	0.00	0.20	0.55	<u>0.90</u>	0.70	0.00	0.00	0.15	<u>0.25</u>	0.05	0.00	0.00	<u>0.00</u>	0.00	0.00
Spline	0.00	0.15	0.55	<u>0.80</u>	0.80	0.00	0.00	0.10	<u>0.20</u>	0.20	0.00	0.00	<u>0.00</u>	0.00	<u>0.00</u>
TV	0.10	0.25	0.45	<u>0.85</u>	0.85	0.00	0.00	0.15	<u>0.40</u>	0.20	0.00	0.00	<u>0.00</u>	0.00	0.00
Weak	0.00	0.15	<u>0.75</u>	0.90	<u>1.00</u>	0.00	0.00	0.10	0.55	<u>0.65</u>	0.00	0.00	0.00	0.05	<u>0.10</u>

TABLE VI

Mean coefficient error on the 20 test seeds at 5% noise for every threshold of the grid; marks as in Table V.

λ	5% noise				
	0.05	0.1	0.2	0.4	0.6
FD	<u>0.1354</u>	<u>0.0469</u>	0.0489	0.0603	0.0711
SG	0.1455	0.0766	0.0745	<u>0.0772</u>	0.1092
Spline	0.1046	0.0597	0.0595	<u>0.0636</u>	0.0793
TV	0.1195	0.0462	0.0470	<u>0.0536</u>	0.0686
Weak	0.0486	0.0211	0.0235	0.0198	<u>0.0337</u>

TABLE VII

Weak-form width sweep at 2% (left three columns) and 5% (right three) noise; $\dagger$ tuned width (main runs).

Width	supp.	coef.err	FP/FN	supp.	coef.err	FP/FN
0.3	1.00	0.0029	0.00/0.00	0.65	0.0276	0.25/0.20
0.6 $\dagger$	1.00	0.0035	0.00/0.00	0.65	0.0337	0.20/0.25
1.0	1.00	0.0058	0.00/0.00	0.45	0.1339	0.40/0.45
1.5	0.60	0.0290	0.30/0.10	0.10	0.0915	0.70/0.75
2.5	0.35	0.0956	0.45/0.70	0.10	0.2379	0.90/1.25
4.0	0.15	0.2117	1.60/1.10	0.00	0.4187	2.90/1.10

each baseline. These conclusions hold for one simulated system, 20 trials per cell and oracle-assisted tuning.

References

- [1] S. L. Brunton, J. L. Proctor, and J. N. Kutz, “Discovering governing equations from data: Sparse identification of nonlinear dynamical systems,” arXiv preprint arXiv:1509.03580, 2015. [Online]. Available: <https://arxiv.org/abs/1509.03580>
- [2] S. H. Rudy, S. L. Brunton, J. L. Proctor, and J. N. Kutz, “Data-driven discovery of partial differential equations,” arXiv preprint arXiv:1609.06401, 2016. [Online]. Available: <https://arxiv.org/abs/1609.06401>
- [3] B. M. de Silva, K. Champion, M. Quade, J.-C. Loiseau, J. N. Kutz, and S. L. Brunton, “PySINDy: A Python package for the Sparse Identification of Nonlinear Dynamics from Data,” arXiv preprint arXiv:2004.08424, 2020. [Online]. Available: <https://arxiv.org/abs/2004.08424>
- [4] H. Schaeffer and S. G. McCalla, “Sparse model selection via integral terms,” Physical Review E, 2017. [Online]. Available: <https://doi.org/10.1103/PhysRevE.96.023302>
- [5] D. A. Messenger and D. M. Bortz, “Weak SINDy: Galerkin-Based Data-Driven Model Selection,” Multiscale Modeling & Simulation, 2021. [Online]. Available: <https://doi.org/10.1137/20M1343166>
- [6] —, “Weak SINDy For Partial Differential Equations,” arXiv preprint arXiv:2007.02848, 2020. [Online]. Available: <https://arxiv.org/abs/2007.02848>
- [7] P. A. K. Reinbold, D. R. Gurevich, and R. O. Grigoriev, “Using noisy or incomplete data to discover models of spatiotemporal dynamics,” Physical Review E, 2020. [Online]. Available: <https://doi.org/10.1103/PhysRevE.101.010203>
- [8] A. Savitzky and M. J. E. Golay, “Smoothing and Differentiation of Data by Simplified Least Squares Procedures,” Analytical Chemistry, 1964. [Online]. Available: <https://doi.org/10.1021/ac60214a047>
- [9] R. Chartrand, “Numerical Differentiation of Noisy, Nonsmooth Data,” ISRN Applied Mathematics, 2011. [Online]. Available: <https://doi.org/10.5402/2011/164564>
- [10] E. N. Lorenz, “Deterministic Nonperiodic Flow,” Journal of the Atmospheric Sciences, 1963. [Online]. Available: [https://doi.org/10.1175/1520-0469\(1963\)020<0130:DNF>2.0.CO;2](https://doi.org/10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2)
- [11] K. Kaheman, J. N. Kutz, and S. L. Brunton, “SINDy-PI: A Robust Algorithm for Parallel Implicit Sparse Identification of Nonlinear Dynamics,” arXiv preprint arXiv:2004.02322, 2020. [Online]. Available: <https://arxiv.org/abs/2004.02322>
- [12] U. Fasel, J. N. Kutz, B. W. Brunton, and S. L. Brunton, “Ensemble-SINDy: Robust sparse model discovery in the low-data, high-noise limit, with active learning and control,” arXiv preprint arXiv:2111.10992, 2021. [Online]. Available: <https://arxiv.org/abs/2111.10992>
- [13] F. van Breugel, J. N. Kutz, and B. W. Brunton, “Numerical differentiation of noisy data: A unifying multi-objective optimization framework,” arXiv preprint arXiv:2009.01911, 2020. [Online]. Available: <https://arxiv.org/abs/2009.01911>